

Simple Linear Regression — Salary Data

From variables to estimators to a prediction

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<https://github.com/HelloJJJie/Assignment-Forecast-and-Analysis-of-Regression.git>

A dataset of $n = 30$ employees — can salary be described as a linear function of experience?

1. The variables and the question

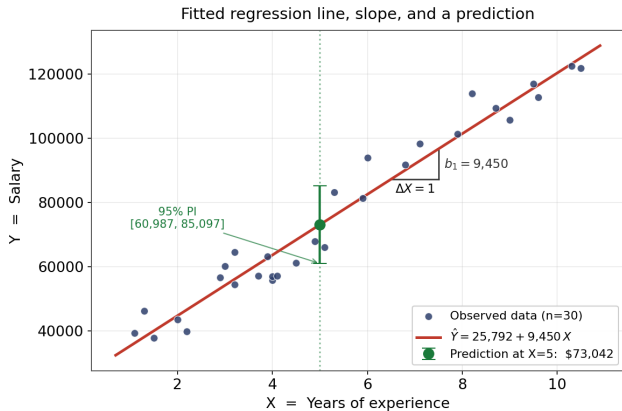
The data. A sample of $n = 30$ employees, with two measured variables:

- $X = \text{years of experience}$
(the *predictor*, ranging 1.1–10.5 years)
- $Y = \text{annual salary}$
(the *response*, what we want to explain)

The model. We assume the mean salary is a straight line in experience,

$$E\{Y\} = \beta_0 + \beta_1 X,$$

and estimate the line from the data by **least squares**. The fitted slope



2. The estimators — reading the fitted line

Least squares chooses b_0, b_1 to minimise $\sum(Y_i - \hat{Y}_i)^2$, giving the fitted regression function

$$\hat{Y} = \underbrace{25,792}_{b_0} + \underbrace{9,450}_{b_1} X$$

Estimator	Value	Std. error
Intercept b_0	25,792	2,273
Slope b_1	9,450	379
Residual spread	$s = \sqrt{MSE} = 5,788$	
Fit quality	$R^2 = 0.957$ ($r = 0.978$)	

- **Slope b_1 :** salary rises \$9,450 per year of experience — and it is clearly non-zero ($t^* = 24.9$, $p \approx 10^{-20}$).
- **Intercept b_0 :** a mathematical anchor; $X = 0$ is outside the data range, so it is *not* a “starting salary”.
- $R^2 = 0.957$: experience accounts for 95.7% of the variation in salary.